Mining Hard Negatives for Contrastive Learning of Brazilian Legal Embeddings

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Abstract. The growing volume of judicial proceedings in Brazil increases the demand for information retrieval methods capable of locating relevant precedents quickly and reliably, a scenario further exacerbated by the scarcity of annotated data in the Brazilian legal domain. This study investigates hard-negative mining as a strategy for contrastive training of Brazilian legal embeddings. Based on appellate decisions from the Brazilian State Court of Maranhão (TJMA), 212,633 training triplets were constructed, each consisting of a headnote (anchor), a chunk from the full text (positive document), and five hard negatives selected based on cosine similarity scores between 0.60 and 0.85 with respect to the query and drawn from distinct cases. These data were used for contrastive fine-tuning of Qwen3-Embedding-4B, resulting in BumbaLM-Embedding-4B. In internal validation, using 9,058 queries, the model achieved an nDCG@10 of 0.747, an MRR@10 of 0.710, and an Accuracy@10 of 86.59% at the best checkpoint, outperforming the untuned base model. To assess generalization to a legal subdomain distinct from that used for training, the model was compared with the BM25 lexical baseline on the BR-TaxQA-R dataset, which focuses on questions and answers concerning Individual Income Tax. BumbaLM-Embedding-4B outperformed BM25 on the evaluated metrics, achieving an nDCG@10 of 66.54% versus 57.13% (a relative gain of 16.5%) and a Recall@10 of 77.81% versus 68.17%. The results indicate that hard-negative mining provides more informative contrastive examples for the specialization of legal embeddings, contributing to retrieval performance both in the source domain and in a distinct legal subdomain.

Keywords. Legal Information Retrieval, Text Embeddings, Hard-negative Mining, Brazilian Case Law, Contrastive Learning

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1. Introduction

The increasing digitalization of the legal sector has expanded the volume of documents available in judicial systems and, consequently, the demand for methods capable of locating relevant information quickly and reliably. In this context, information retrieval techniques and semantic document representation have assumed an important role in organizing and accessing legal knowledge. In Brazil, this demand is particularly relevant given the scale of the judicial case backlog. According to data from the National Council of Justice [1], the Brazilian Judiciary ended 2025 with a massive backlog of 75.5 million pending cases and recorded more than 40.9 million new cases, the highest volume ever recorded. Despite the reduction in the number of pending cases compared with the previous year, these figures highlight the substantial demand for judicial services and make document organization and retrieval a complex challenge. In this scenario, Natural Language Processing (NLP) techniques have increasingly been employed to address problems in the legal domain. These techniques enable the conversion of raw texts from various Brazilian courts into computational representations, seeking to capture specific characteristics of Brazilian legal language [2].

In this context, legal research is a recurring task in the daily activities of judges, attorneys, and other legal professionals, requiring considerable time to locate precedents that support legal analyses and decisions [3]. With the advancement of systems based on Retrieval-Augmented Generation (RAG) architectures, the possibilities for accessing information contained in large document collections have expanded. However, it has become evident that the quality of generated responses is closely related to the retrieval stage, since the selection of irrelevant passages or the omission of essential information compromises the final result [4]. Therefore, it is necessary to employ embedding models capable of identifying relationships between queries and documents not only through lexical matching but also through semantic similarity.

The training of retrieval models depends on the availability of pairs of queries and relevant documents. In the Brazilian legal domain, manually constructing such data requires considerable time and specialized domain knowledge, making available annotated data scarce, difficult to access, and limited to a small number of annotated queries [5]. Another limitation is the limited availability of open-source code and validated datasets, which constrains the reproducibility of studies [6]. However, legal documents exhibit structures and internal relationships that can provide relevance signals without requiring manual annotation for each example. Sections such as headnotes, reports, and legal reasoning, for example, establish semantic relationships with the full text of decisions and can be exploited as supervision signals for constructing positive pairs. Accordingly, contrastive learning strategies use instances composed of a query, a positive document, and one or more negative documents. This structure is intended to maximize the learning signal provided by each example. From these instances, contrastive loss functions are applied to bring the representations of the query and relevant document closer while simultaneously pushing them away from the representations of irrelevant documents [7,8].

However, the effectiveness of this training depends on the quality of the negative examples. When selected randomly, they tend to be easily distinguishable from the positive document, providing limited learning signal for semantic relationships [9]. To mitigate this issue, hard negatives are employed, consisting of documents that are semantically close to the query in the representation space but are not relevant to it. In the context of

court decisions, such examples make the contrast more informative by forcing the model to distinguish between similar documents. However, their selection must be performed carefully, as relevant documents may be incorrectly used as negatives, resulting in false negatives [10].

In the Brazilian context, recent studies have adapted embeddings to the national legal domain, indicating that specialized models tend to outperform general-purpose architectures on retrieval tasks, highlighting the importance of alignment between training data and the characteristics of legal texts [11]. Accordingly, this study analyzes the use of hard-negative mining in the contrastive training of embedding models for Brazilian legal texts. To this end, triplets were constructed from the relationship between the headnote and the full text of court decisions, complemented by semantically similar negatives originating from distinct cases. These data were used to fine-tune Qwen3-Embedding-4B [12], resulting in BumbaLM-Embedding-4B, a model adapted to the semantic characteristics of Brazilian case law. The remainder of this paper is organized as follows: Section 2 presents related work. Section 3 details the methodology, including data preparation and model training. Section 4 discusses the experimental results and metric analyses. Finally, Section 5 presents the concluding remarks and perspectives for future work.

2. Related Work

Information retrieval in the legal domain presents specific technical challenges. The characteristics that make a vector representation model effective for general information retrieval tasks are not necessarily the same as those required in the legal domain [13]. In legal documents, the relevance between a query and a court decision may not depend on the repetition of the same terms, since legal language is characterized by technical terms and low redundancy, with legal concepts being used in ways that differ from common language [3]. Lexical methods such as BM25 [14] are subject to the vocabulary mismatch problem, in which a query contains terms that are semantically equivalent to those in a relevant document but lexically distinct, making it difficult to retrieve relevant documents [15]. The use of generic or low-quality embeddings can compromise retrieval results, leading to inaccurate responses [13]. To overcome these limitations, dense retrieval approaches represent queries and documents in a continuous vector space, allowing proximity between embeddings to capture semantic relationships [16]. However, this approach depends on the model's ability to capture the characteristics of legal texts.

Recent studies have evaluated the impact of language models on retrieval and legal document understanding tasks in Portuguese. The study by [17] directly addressed semantic retrieval in long legal documents from Brazil. In that study, the authors evaluated five variants of BERT models on a corpus of court decisions, using document segmentation and vector retrieval with Elasticsearch. The results showed that BumbaBERT, which was pretrained specifically on Brazilian legal texts, achieved superior performance, demonstrating the relevance of domain specialization for this task. In addition, [11] analyzed specialized embedding models for legal question answering in Brazilian Portuguese through fine-tuning. The results indicate that domain-adapted models can outperform general-purpose models on retrieval tasks evaluated using metrics such as MRR@10 and Recall@10. Complementarily, the study by [18] proposes an evaluation dataset focused on case law retrieval in Portuguese, enabling the comparison of lexical

and dense methods across several metrics. Taken together, these studies indicate that retrieval performance depends not only on the model architecture but also on the alignment between the training data and the characteristics of the domain.

In addition to domain specialization, training strategies are another important factor in optimizing the learning of embedding models. The study by [8] employed bi-encoder architectures to generate representations that can be compared through vector similarity, thereby enabling efficient large-scale semantic retrieval. Complementarily, [7] demonstrated the effectiveness of training dense retrievers with positive and negative pairs, including hard negatives, to bring queries closer to relevant passages while distancing them from irrelevant documents. Large Language Models (LLMs) have also been widely explored for textual embeddings [19]. One commonly used model is Qwen3-Embedding [12], proposed for multilingual embedding and reranking tasks. In this context, the study by [16] evaluated the adaptation of Qwen3-Embedding-4B for legal retrieval, using different training strategies for legal-domain specialization and general-purpose tasks. This evaluation relies on the JUÁ benchmark, which provides common metrics for retrieval in Brazilian legal texts [5]. Finally, although hard negatives provide more informative examples for contrastive training, their selection must be performed carefully, as relevant documents may be incorrectly classified as negatives [10].

In contrast to the studies discussed above, the present study fine-tunes Qwen3-Embedding-4B with a focus on Brazilian case law. The proposed strategy is based on the construction of training triplets and the application of hard-negative mining to court decisions from a Brazilian state court. Thus, the objective is to empirically evaluate the gains provided by this method in contrastive training and to develop a model for generating embeddings specialized in the legal domain.

3. Methodology

To guide the development of the proposed model, the CRISP-DM (Cross-Industry Standard Process for Data Mining) model [20] was adopted, structuring the process into six stages: 1) business understanding; 2) data understanding; 3) data preparation; 4) modeling; 5) evaluation; and 6) deployment. In this context, BumbaLM-Embedding-4B serves as an embedding model capable of supporting NLP tasks, developed for semantic retrieval of legal texts. For its construction, Qwen3-Embedding-4B [12] was used as the base model, since it is already trained to project queries and documents into the same vector space and has been evaluated on MTEB [21] retrieval tasks. Thus, fine-tuning specializes the model for the legal domain, and the quality of the resulting representations is evaluated through information retrieval.

3.1. Business and Data Understanding

The objective of this study was to obtain vector representations of Brazilian court decisions that preserve semantic proximity between legally related texts. Although state courts account for the majority of pending cases in Brazil, encompassing more than 58 million cases according to the Justice in Numbers report [1], each court exhibits particularities in vocabulary and structural organization. In this context, court decisions from TJMA were used to develop a specialized model.

The dataset used in this study was provided by TJMA and contained approximately 100,000 records. To structure the training instances, the documents were standardized around three fundamental fields: case number, headnote, and full text. The headnote summarizes the legal holding and serves as the query in retrieval tasks, whereas the full text constitutes the complete document to be retrieved [22], thus forming contextual pairs. However, the extracted texts exhibited variations in length, text markup, and the inclusion of metadata within the body of the decision, requiring preprocessing of the dataset.

3.2. Data Preparation

The extraction of legal documents frequently results in a noisy corpus, containing formatting remnants and residual structures that can hinder the learning process of language models [23]. Therefore, to obtain a usable and high-quality sample for the subsequent hard-negative mining stages, preprocessing was performed, including normalization and the removal of HyperText Markup Language (HTML), invisible characters, and multiple whitespace characters. In addition, empty records or records containing insufficient text were discarded. Duplicate documents were also removed based on the case number, resulting in a sample of approximately 60,000 valid pairs consisting of the headnote and full text as the anchor and positive document, respectively. The pairs were then divided into training and validation sets for training and evaluating BumbaLM-Embedding-4B.

In retrieval tasks, compressing long documents into a single fixed-size vector representation creates an information bottleneck that tends to dilute semantic nuances and hinder the identification of the relevant passage [24]. In the legal domain, texts exhibit a sequential structure, and simply truncating such documents can result in substantial context loss, omitting important central arguments. Therefore, to address this issue, a chunking strategy was adopted to handle long legal documents [17]. Each full text was divided into windows of 768 tokens, using an overlap of 128 tokens to preserve part of the context at chunk boundaries, while the headnote remained intact as the query.

3.3. Modeling

The modeling stage involved constructing contrastive training instances from the pairs generated during data preparation. In retrieval models, this approach seeks to bring queries and relevant documents closer together in the vector space while pushing irrelevant documents farther away [7]. However, in specific domains such as the legal domain, retrieval is affected by semantically similar but contextually irrelevant candidates due to terminological overlap and semantic mismatch [25].

Distinct documents may exhibit similar vocabulary without corresponding to the same relevance relationship [18]. Accordingly, a hard-negative mining strategy was employed, selecting documents that exhibit high semantic similarity to the anchor but belong to distinct cases. These negatives must be selected carefully, as randomly sampled negative examples may be trivial, whereas semantically similar negatives provide stronger learning signals for distinguishing between documents [10]. Thus, for each headnote, the chunks originating from other cases were initially ranked according to cosine similarity. To construct the hard negatives, five candidates with similarity scores between 0.60 and 0.85 with respect to the query were selected. The headnote, positive

chunk, and five negative examples were then combined to form training instances in triplet format. At the end of this stage, the training set comprised 212,633 samples, each consisting of a headnote, one positive chunk from the full text, and five hard negatives originating from other cases.

For the fine-tuning stage using the generated instances, Qwen3-Embedding-4B[2] was used as the base model. To adapt the model to the legal domain, the queries were prefixed with the instruction in Portuguese "Instruct: Dado um trecho de um documento jurídico brasileiro, recupere o trecho mais relevante semanticamente.", while the positive and negative documents were processed without additional instructions. This formatting follows the scheme recommended for embedding models based on Large Language Models (LLMs) [12]. Training was implemented using the SentenceTransformers framework [8], with the Cached Multiple Negatives Ranking Loss function. This loss function considers the multiple hard negatives associated with each query, in addition to the documents from the other instances in the training batch. Last-token pooling and normalization of the resulting vectors were employed. To enable updating all parameters of the Transformer, memory consumption was optimized using gradient checkpointing and bfloat16 precision. Training was conducted for one epoch, with a batch size of 16 instances per GPU, gradient accumulation over two steps, a learning rate of 2×10^{-5}, and a warmup over the first 6% of the training steps.

3.4. Evaluation and Deployment

In legal Portuguese, general-purpose embeddings tend to underrepresent domain-specific vocabulary and context, which can impair semantic retrieval in RAG pipelines [11]. Therefore, the evaluation stage was structured to determine whether fine-tuning improved the representations for retrieval. The validation process used the headnotes as queries to rank the chunks in the corresponding corpus. The evaluation set consisted of 9,058 positive documents, distributed at a ratio of one chunk per case, ensuring that each search was associated with a single relevant document.

The primary metric was nDCG@10, which measures ranking quality among the top ten results [26]. Accuracy, Precision, Recall, MRR, and MAP were also reported, as these metrics are commonly used for this type of task, as in [18]. The baseline was evaluated using the same queries, given that it is already a retrieval embedding model evaluated in legal benchmarks that employ ranking metrics [5]. In terms of deployment, the resulting artifact is the BumbaLM-Embedding-4B embedding model, specialized in the legal domain. In addition to enabling the evaluation of contrastive training based on hard negatives, the resulting trained and evaluated representation can be made available to TJMA to support NLP tasks over local case law collections.

To evaluate the performance of the proposed model in a legal scenario distinct from the dataset used for training, BR-TaxQA-R [27] was used, a dataset derived from materials provided by the Brazilian Federal Revenue Service concerning Individual Income Tax (IRPF). The dataset contains 715 questions and their corresponding answers, formulated as frequently asked questions (FAQ) in the tax-law domain. Some questions have explicit relationships with other questions, allowing different levels of relevance to be considered during retrieval. In this context, the answer directly associated with the query was defined as the primary positive document, with a relevance score of 2, while answers

[2] `https://huggingface.co/Qwen/Qwen3-Embedding-4B`

linked to related questions were considered secondary positives, with a score of 1. This configuration enables evaluation with multiple relevance levels, which is more appropriate for scenarios in which different documents exhibit varying degrees of relevance to the query. On this dataset, BumbaLM-Embedding-4B was compared with BM25, used as a lexical baseline, to evaluate the model's ability to retrieve relevant documents under different representation and query-document matching strategies.

4. Results and Analysis

To assess the impact of fine-tuning on the vector representations of the appellate decisions, the progression of ranking metrics across the model update steps was analyzed. This monitoring enables the identification of the optimal convergence point and the evaluation of contrastive training stability. The primary metric for checkpoint selection was the Normalized Discounted Cumulative Gain [26], as it captures the ranking quality at the top positions. Figure 1 details the model's behavior.

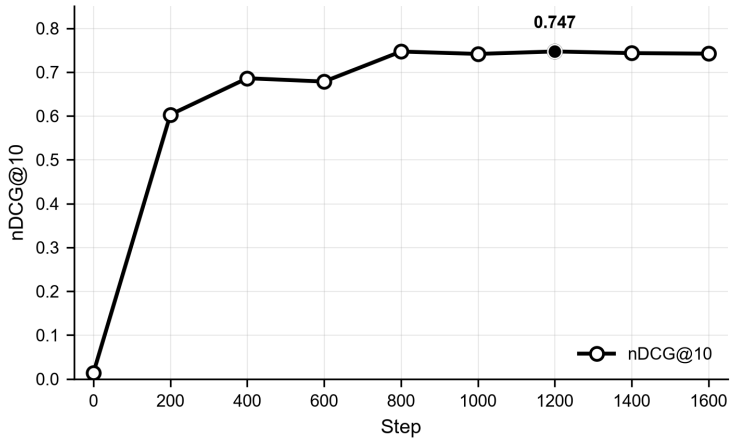


Figure 1. Evolution of nDCG@10 on the validation set throughout the training steps.

Initially, a marked improvement is observed in the first few hundred steps, demonstrating the assimilation of the semantic characteristics of the case law. As training progresses, the learning curve stabilizes, indicating that the model has consolidated its ability to distinguish relevant documents from hard negatives. Peak performance on the validation set is achieved at checkpoint 1200, when the nDCG@10 metric reaches a value of 0.747, representing the optimal generalization state prior to the end of training. Additionally, Figure 2 presents the MRR@10 and Accuracy@{1,10} at the same evaluation points.

Based on the observation of the metrics, it is noted that at checkpoint 1200, the model achieved an MRR@10 $= 0.710$, Accuracy@1 $= 63.48\%$, and Accuracy@10 $= 86.59\%$. These results demonstrate that, in addition to the overall ranking improvement, the relevant chunk comes to occupy the top positions of the list for the vast majority of queries. As a starting point for evaluation, the Qwen3-Embedding-4B base-

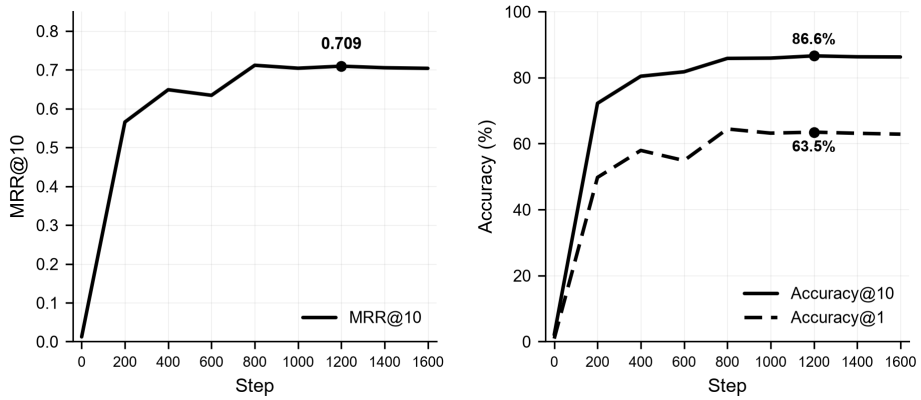


Figure 2. Evolution of MRR@10 and Accuracy@k ($k \in \{1, 10\}$) on the validation set throughout the training steps.

line [12], without fine-tuning, recorded minimal precision and ranking metrics under the same protocol. Table 1 presents the complete set of metrics at the training evaluation points, enabling the observation of their evolution.

Table 1. Information retrieval evaluation results on the validation set throughout the training of the BumbaLM-Embedding-4B model.

Step	Acc@1	Acc@10	MRR@10	nDCG@10	MAP@100
0	0.91%	2.06%	0.012	0.014	0.013
200	49.76%	72.21%	0.566	0.603	0.572
400	57.94%	80.43%	0.649	0.686	0.655
600	54.87%	81.76%	0.634	0.678	0.641
800	64.44%	85.82%	0.708	0.747	0.717
1000	63.17%	85.91%	0.704	0.741	0.710
1200	**63.48%**	**86.59%**	**0.709**	**0.747**	**0.715**
1400	63.13%	86.31%	0.705	0.743	0.711
1600	62.86%	86.26%	0.704	0.742	0.709

As demonstrated in the dense retrieval literature [9], overly easy negatives tend to produce little learning signal, as the model separates them from the query without requiring fine-grained semantic distinctions. Consequently, by using five candidates with high semantic similarity originating from different legal proceedings, the contrastive training resulted in a substantial enhancement of BumbaLM-Embedding-4B's ability to retrieve the correct excerpt based on the headnote. In addition to the internal evaluation on the court's own documents, when evaluating the model against the lexical baseline BM25 on the BR-TaxQA-R dataset [27], a consistent gain in the evaluated metrics was observed, even outside the internal training set. Table 2 consolidates the metrics achieved by the two approaches.

In terms of nDCG@10, BM25 achieved 57.13% compared to 66.54% for BumbaLM-Embedding-4B, representing a difference of +9.42 percentage points in favor of the dense model. The most significant relative gain occurs at nDCG@10. Table 3 details the behavior of nDCG@k and Recall@k across the evaluated cutoffs. The advantage of

Table 2. Comparison of retrieval performance between BM25 and BumbaLM-Embedding-4B on the BR-TaxQA-R dataset ($k = 10$). Bold values indicate the best performance, and Δ represents the absolute percentage point (p.p.) difference.

Metric	**BM25**	**BumbaLM-Embedding-4B**	**Δ (p.p.)**	**Δ (%)**
nDCG@10	57.13	**66.54**	+9.42	+16.48
MAP@10	49.53	**57.67**	+8.14	+16.44
MRR@10	58.86	**67.13**	+8.27	+14.06
Recall@10	68.17	**77.81**	+9.64	+14.14
P@10	10.21	**10.88**	+0.67	+6.57

BumbaLM-Embedding-4B is maintained across all cutoffs, including the coverage of relevant items: at Recall@100, the dense model reaches 94.43% compared to 90.02% for BM25.

Table 3. Comparison of nDCG@k and Recall@k between BM25 and BumbaLM-Embedding-4B across different cutoff values (k) on the BR-TaxQA-R dataset. Bold values indicate the best performance, and Δ represents the absolute percentage point (p.p.) difference.

	nDCG@k			**Recall@k**		
k	BM25	BumbaLM	Δ (p.p.)	BM25	BumbaLM	Δ (p.p.)
1	47.34	**55.32**	+7.97	38.02	**45.58**	+7.56
3	51.81	**60.84**	+9.03	52.67	**61.96**	+9.28
5	54.26	**63.81**	+9.55	59.44	**69.75**	+10.31
10	57.13	**66.54**	+9.42	68.17	**77.81**	+9.64
100	61.98	**70.43**	+8.45	90.02	**94.43**	+4.41
1000	63.32	**71.20**	+7.88	99.72	**100.00**	+0.28

The results indicate that BumbaLM-Embedding-4B consistently outperforms BM25 in a legal subdomain distinct from that used during fine-tuning, both in ranking quality at the top positions and in the ability to retrieve relevant items. In conjunction with the internal validation on the corpus of appellate decisions, these findings demonstrate that contrastive specialization based on hard negatives enhances semantic representation, optimizing legal retrieval even beyond the specific training scenario. Expanding this analysis to include other dense baselines emerges as a promising direction for future work. The repository containing the obtained results, as well as additional resources that may facilitate reproducibility, will be available at `https://github.com/alekxDias/BumbaLM-Embedding-4B`.

5. Conclusion

This work evaluated the impact of contrastive fine-tuning with hard negative mining on the quality of embeddings for the semantic retrieval of appellate decisions from the TJMA. Based on contextual pairs, combined with the technique of dividing the full text into chunks and the subsequent selection of five hard negatives, full fine-tuning was performed, resulting in BumbaLM-Embedding-4B. In the validation protocol with 9,058 queries and one relevant chunk per case, the model achieved an nDCG@10 of 0.747,

Accuracy@10 of 86.59%, and MRR@10 of 0.710. Furthermore, when evaluated on the BR-TaxQA-R legal dataset, BumbaLM-Embedding-4B consistently outperformed the lexical BM25 baseline across all metrics, with a relative gain of 16.5% in nDCG@10 and 14.1% in Recall@10. These results demonstrated that specialization on the pairs formed by the headnote and the chunk from the legal text corpus of the TJMA significantly alters the ranking quality, reinforcing the premise that legal domain specialization consistently outperforms the use of generic representations.

Despite the observed gains, this work presents limitations that should be considered when interpreting the results. The model was specialized exclusively on appellate decisions from the TJMA, which may restrict direct generalization to other courts or judicial instances, whose lexical and structural particularities may differ. Another factor to consider is that, although the evaluation on BR-TaxQA-R presented evidence of the model's generalization to a distinct domain, this comparison was initially performed against a lexical baseline, without comparison to other general-purpose models or those specialized in the legal domain. Furthermore, the evaluation focused on the information retrieval task, chosen because it enables a direct measurement of the quality of the vector representation produced by the fine-tuning. However, performance on this specific task does not limit the application of the generated embeddings, which can be employed in other legal NLP tasks, such as appellate decision classification, decision clustering, or similarity detection, which are scenarios that were not explored in this work.

As future work, the intention is to extend the hard negative mining methodology to corpora from other state and federal courts, evaluating the impact of institutional diversity on the quality of the representations. It is also planned to investigate more refined mining strategies and the combination of multiple contrastive loss functions, in order to mitigate the risk of false negatives. Furthermore, the BumbaLM-Embedding-4B model is intended to be made available to the TJMA to support tasks on the local corpus. Finally, it is expected that the specialized model can be utilized to integrate internal systems across multiple applications.

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